Name: Jose Salinas
Course: CS-6370
Professor: Meilin Liu
Date: 11/9/2026
Project Name: Project 3 Reduce and Scan

Report:

Both reduce and scan program compile. All project files have course number, professor name, data, and the project name. The README will instruct how to compile and use the reduce and scan program. Executing the program will execute respective calculations on CPU and GPU and print the results on the screen. Both reduction and scan executions results are correct for all mandatory executions. Scan does not work with 4096 * 4096 matrix. Reduce is able to be used with 4096 * 4096 array. Both will print the first 20 elements of the array, when able to. Both programs will compare the GPU scanned and reduced computed arrays with the CPU computed scan and reduce and will print “Test PASSED” if they match. NOTE: The way I implemented the reduce for 4096 * 4096 I use both input and output pointers doing a double buffer type of implementation.

Reduce

Time	131072 (128*1024) Required for undergraduate students and graduate students (30 points)	1048576 (1024*1024) Required for undergraduate students and graduate students (30 points)	Bonus: 16777216 (4096*4096) 10 points
CPU computation time	0.420000	0.106144	0.526400
GPU computation time	0.090048	2.868000	43.584999
GPU memory transfer time			

Scan:

Time	2048 (2*1024) Required for undergraduate students and graduate students (40 points)	1048576 (1024*1024) Required for graduate students (40 points)	Bonus: 16777216 (4096*4096) Optional: 20 points
CPU computation time	0.011000	4.558000	
GPU computation time	0.052256	0.151328	

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GPU memory transfer time	0.23427	2.980640	

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```
scan.cu:161:24: warning: non-static data member initializers only ava
d=gnu++11
   float milliseconds = 0;
                        ^
[w859jas@smgpu01 reduce]$ ./scan
Would you like to input factored length (NxM)? [y/n] y
Input width: 2
Input height: 1024
Length will be 2048
Current Input
71 41 -62 47 -42 58 84 31 -75 -74 68 84 6 26 58 -44 0 49 84 80
THREAD_BLOCK 1024
CUR GRID SIZE 1
GPU time taken 0.052256 milliseconds
CPU time taken 0.011000 milliseconds
GPU Memory time taken 0.234272 milliseconds
CPU Output
71 112 50 97 55 113 197 228 153 79 147 231 237 263 321 277 277 326 41
GPU Output
71 112 50 97 55 113 197 228 153 79 147 231 237 263 321 277 277 326 41
Test PASSED
[w859jas@smgpu01 reduce]$ ./scan
Would you like to input factored length (NxM)? [y/n] y
Input width: 1024
Input height: 1024
Length will be 1048576
Current Input
71 41 -62 47 -42 58 84 31 -75 -74 68 84 6 26 58 -44 0 49 84 80
THREAD_BLOCK 1024
CUR GRID SIZE 512
GPU time taken 0.151328 milliseconds
CPU time taken 4.558000 milliseconds
GPU Memory time taken 2.980640 milliseconds
CPU Output
71 112 50 97 55 113 197 228 153 79 147 231 237 263 321 277 277 326 41
GPU Output
71 112 50 97 55 113 197 228 153 79 147 231 237 263 321 277 277 326 41
Test PASSED
[w859jas@smgpu01 reduce]$
```

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```
[w859jas@smgpu01 reduce]$ ./reduce
Would you like to input factored length (NxM)? [y/n] y
Input width: 128
Input height: 1024
Length will be 131072
Current Input
71 41 -62 47 -42 58 84 31 -75 -74 68 84 6 26 58 -44 0 49 84 80
THREAD_BLOCK_SIZE: 64
Initial Grid Size: 1024
GPU time taken 0.090048 milliseconds
CPU time taken 0.420000 milliseconds
Test PASSED
CPU Output: 4208394, GPU Output: 4208394
[w859jas@smgpu01 reduce]$ ./reduce
Would you like to input factored length (NxM)? [y/n] y
Input width: 1024
Input height: 1024
Length will be 1048576
Current Input
71 41 -62 47 -42 58 84 31 -75 -74 68 84 6 26 58 -44 0 49 84 80
THREAD_BLOCK_SIZE: 64
Initial Grid Size: 8192
GPU time taken 0.106144 milliseconds
CPU time taken 2.868000 milliseconds
Test PASSED
CPU Output: 33666891, GPU Output: 33666891
[w859jas@smgpu01 reduce]$ ./reduce
Would you like to input factored length (NxM)? [y/n] y
Input width: 4096
Input height: 4096
Length will be 16777216
Current Input
71 41 -62 47 -42 58 84 31 -75 -74 68 84 6 26 58 -44 0 49 84 80
THREAD_BLOCK_SIZE: 64
Initial Grid Size: 131072
GPU time taken 0.526400 milliseconds
CPU time taken 43.584999 milliseconds
Test PASSED
CPU Output: 538668663, GPU Output: 538668663
```